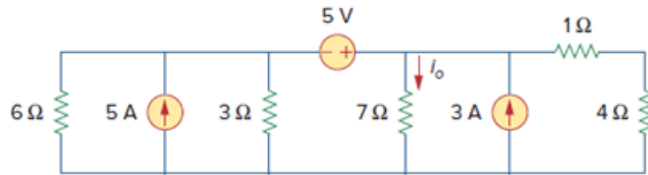


Problem

Find i_o in the circuit of Fig. 4.19 using source transformation.

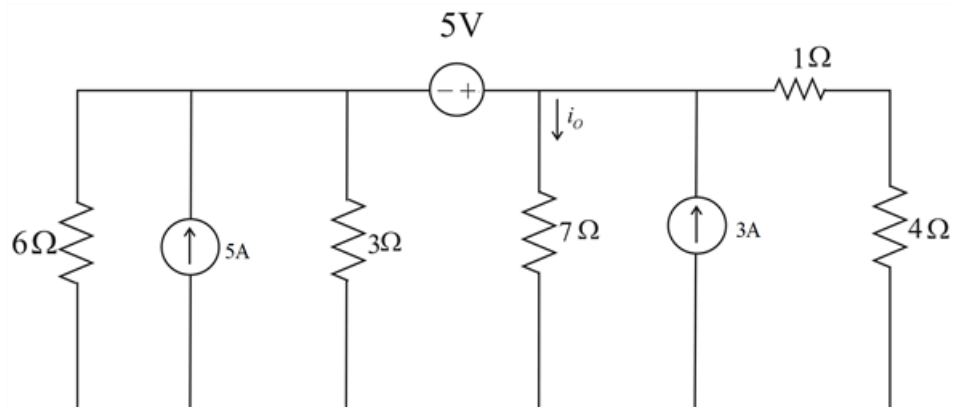
Figure. 4.19:



Step-by-step solution

Step 1 of 7

Given circuit is



Step 2 of 7

From the above circuit $6\Omega, 3\Omega$ are in parallel

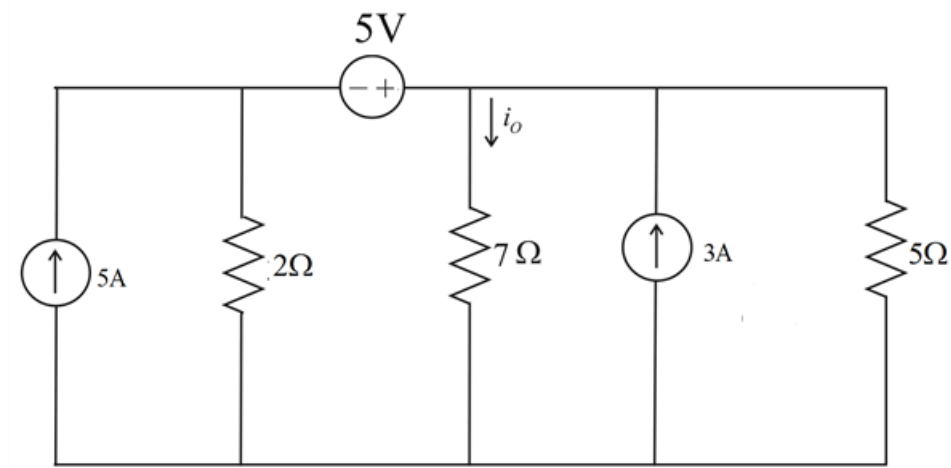
$$\Rightarrow \frac{6 \times 3}{6 + 3}$$

$$\Rightarrow \frac{18}{9}$$

$$\Rightarrow 2\Omega$$

And $1\Omega, 4\Omega$ are in series so the circuit can be drawn as

Step 3 of 7



Step 4 of 7

From source transformation current source in parallel with the resistor can be replaced by voltage source in series with the resistor

So voltage source value is

$$v_1 = 2 \times 5$$

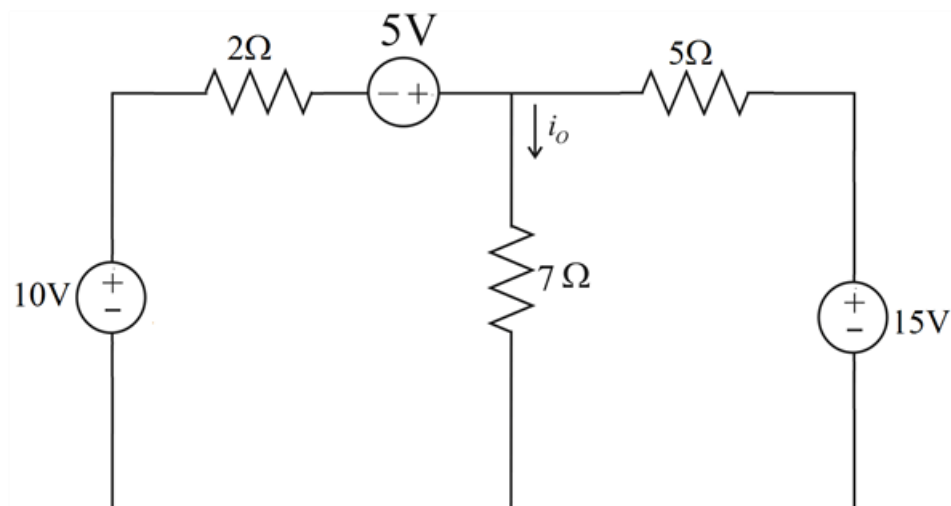
$$v_1 = 10V$$

$$v_2 = 3 \times 5$$

$$v_2 = 15V$$

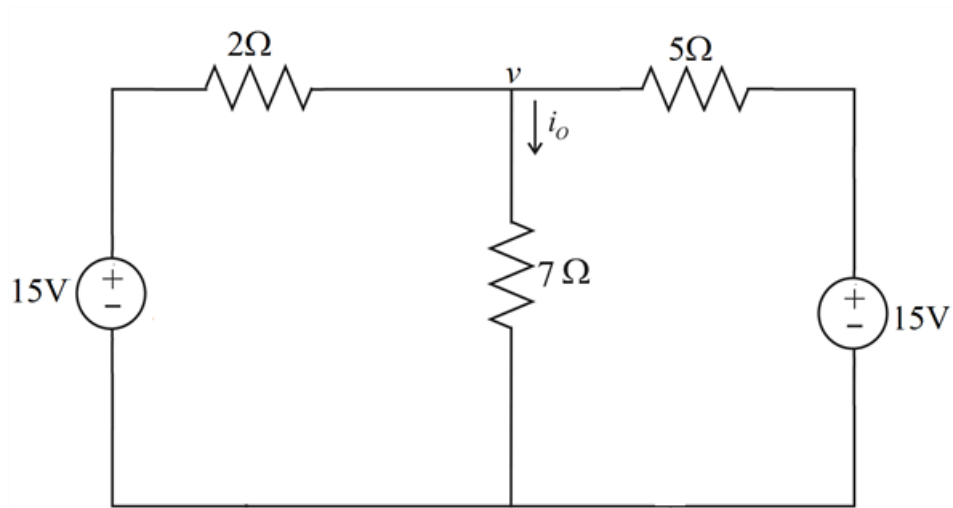
Now the modified circuit is

Step 5 of 7



Step 6 of 7

Circuit can be redrawn as



Step 7 of 7

Applying KCL at top node

$$\frac{v-15}{2} + \frac{v}{7} + \frac{v-15}{5} = 0$$

$$v \left(\frac{1}{2} + \frac{1}{7} + \frac{1}{5} \right) = \frac{15}{2} + \frac{15}{5}$$

$$0.8428v = 10.5$$

$$v = \frac{10.5}{0.8428}$$

$$v = 12.45\text{V}$$

$$i_o = \frac{v}{7}$$

$$i_o = \frac{12.45}{7}$$

$$\boxed{i_o = 1.78\text{A}}$$